



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to PRANAV AGENCY UNIT II

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

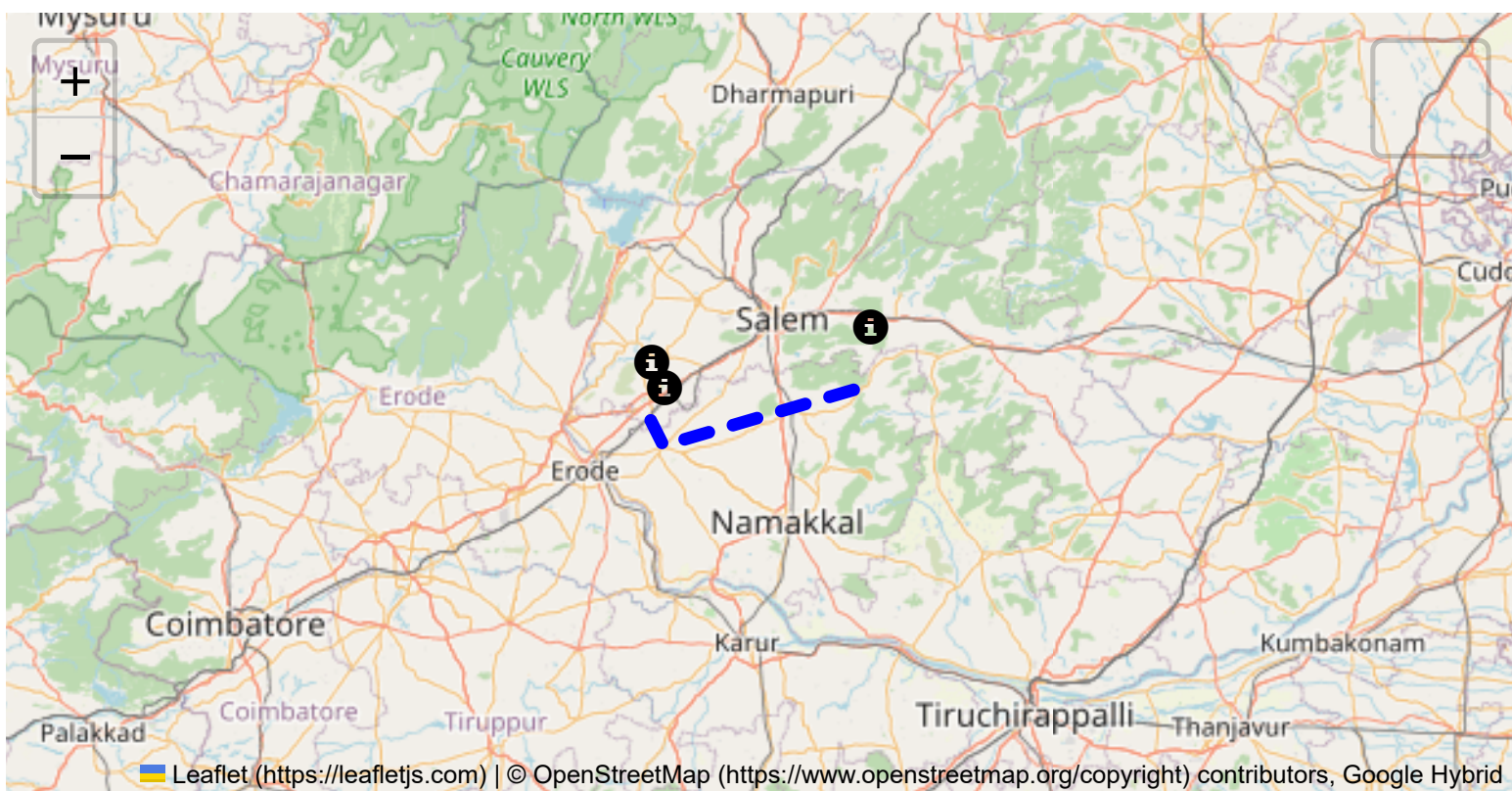
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

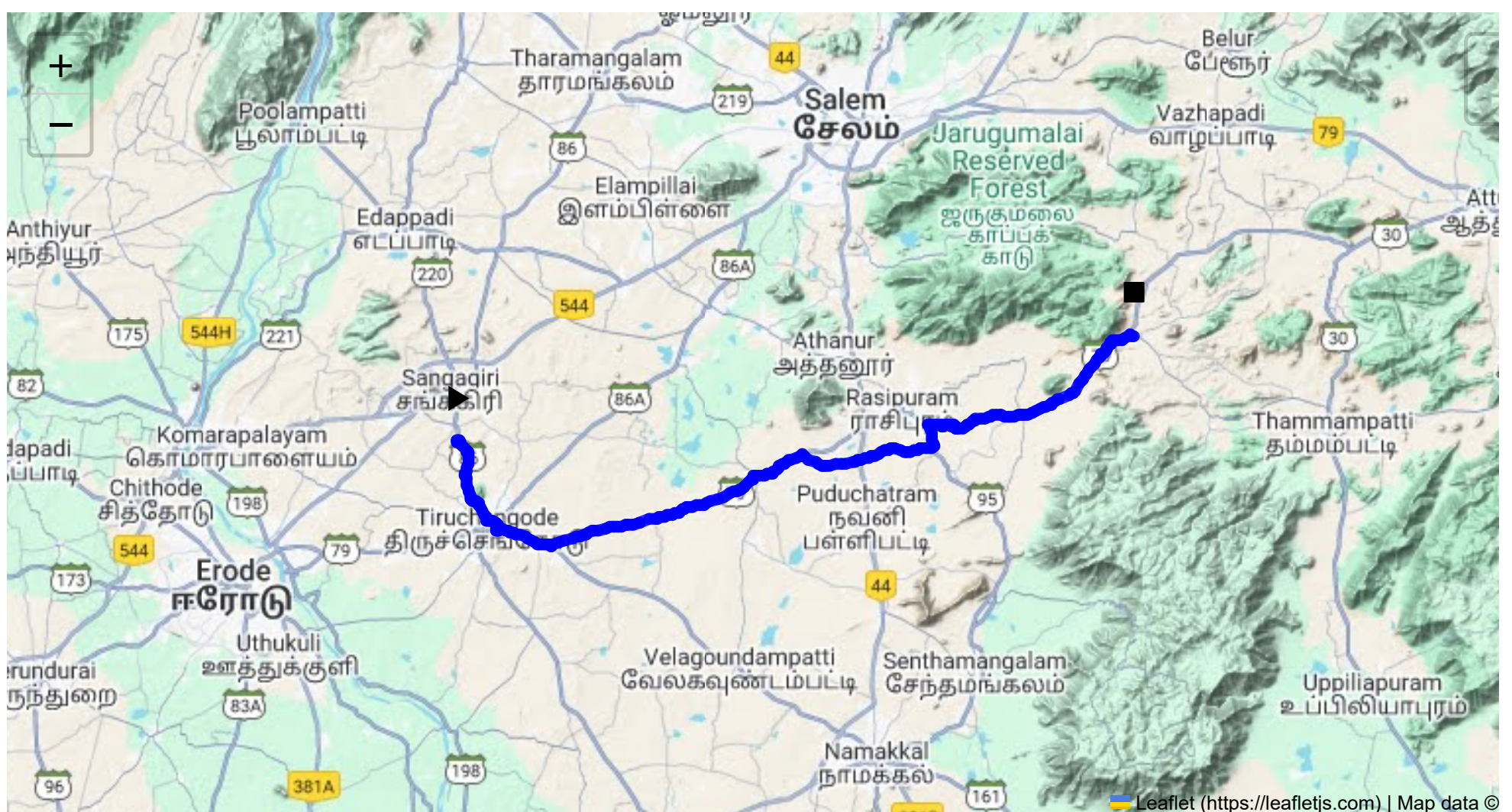
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 65.18 km**  
**Estimated Duration: 1.5 hours**  
**Adjusted Duration (Heavy Vehicle): 1.9 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.5104058, 78.34693408)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

This route begins at CVQF+23W, Sangagiri, and proceeds northeast towards G86W+4RC, Mettala Rd, passing via 20A, Old Salem Rd, MUTHUMANTHOPPU, and MUTHUSELVAM NAGAR, in Tiruchengode. The journey covers around 65.18 kilometers through a mixture of urban and semi-rural environments. This



area has a mix of two-lane roads and smaller local roads, which can impact vehicle movement, especially for heavy vehicles.

## 2. Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate with high temperatures and seasonal monsoons. The primary weather-related hazards include:

- **Monsoon Season (June to September):** Heavy rain can lead to flooding and reduced visibility.
- **Cyclones and Storms:** Can occasionally affect the region during the late monsoon and post-monsoon period (October to December).

## 3. Traffic Patterns

- **Peak Traffic Hours:** Typically from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM.
- **Congestion-Prone Areas:**
  - Urban areas around Tiruchengode.
  - Entry and exit points to larger roads from Old Salem Rd and smaller local roads may experience bottlenecks.

## 4. Road Quality and Infrastructure

- **20A and Old Salem Rd:** Generally maintained but may have sections with potholes and uneven surfaces, especially post-rains.
- **Local Roads:** May have limited signage and narrower stretches, requiring cautious navigation.

## 5. Alternative Routes for Emergencies

- Alternative routes include bypasses through Erode for a longer but potentially quicker route during obstructions.
- Use GPS navigation apps which provide real-time updates on traffic and road conditions to guide the rerouting.

## 6. Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for hazardous materials:

- Restrictions on timings for transport through urban areas.
- Requirement for mandatory permits and adherence to signage.

## 7. Historical Incidents

While there are no recent major incidents involving hazardous materials on this specific route, nearby urban areas have witnessed accidents due to heavy traffic. Drivers should maintain caution, especially in congested zones.

## 8. Environmental Considerations and Sensitive Areas

- Areas near water bodies are sensitive and must be navigated with caution to prevent contamination in case of spills.
- Agricultural land is prevalent, where spillage could harm crops.

## 9. Communication Coverage

- Most urban and semi-urban areas have good cellular coverage, but occasional dead zones might be encountered in rural sections or near dense foliage between Tiruchengode and Sangagiri.

## 10. Estimated Emergency Response Times

- Urban areas: 15–30 minutes.
- Semi-rural portions: 30–45 minutes.
- In remote or less accessible areas, times might increase beyond 45 minutes.

## 11. Overall Summary of Risk Assessment

This route presents moderate risks for transporting hazardous materials, largely due to:

- Possible weather-related disruptions.
- Congestion in urban areas.
- Mixed road quality requiring vigilance.

## Recommendations:

- Plan travel times to avoid peak hours.
- Monitor weather forecasts and road condition alerts.
- Ensure compliance with all regulatory requirements for safety.
- Optimize communication strategies to mitigate potential dead zone issues.

By adhering to these guidelines and remaining alert, truck drivers can safely navigate this route while minimizing potential hazards.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 2 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 3  | Blind Spot | Blind Spot | 11.44029, 77.87544 | 10 KM/Hr    |
| 4  | Turn       | Medium     | 11.38346, 77.89975 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.38302, 77.89995 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.38188, 77.90257 | 15 KM/Hr    |
| 7  | Blind Spot | Blind Spot | 11.37841, 77.90140 | 10 KM/Hr    |
| 8  | Turn       | Medium     | 11.36745, 77.93951 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.41332, 78.08072 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.41463, 78.08067 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.42964, 78.11587 | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.42454, 78.15291 | 30 KM/Hr    |
| 13 | Turn       | High       | 11.43542, 78.20754 | 15 KM/Hr    |
| 14 | Blind Spot | Blind Spot | 11.45100, 78.20385 | 10 KM/Hr    |
| 15 | Turn       | Medium     | 11.44925, 78.22288 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.44813, 78.22305 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.44793, 78.22320 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.49628, 78.32342 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.50497, 78.33079 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.50769, 78.33291 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.50860, 78.33951 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.51180, 78.34422 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital    | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode     | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                      | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 10 | hospital | Government Hospital, Namagiripettai | 11.4569484, 78.265263  | 30 km/h     | Medium     |

## Crowded Spots

|   | type        | name                                      | coordinates               | speed_limit | risk_level |
|---|-------------|-------------------------------------------|---------------------------|-------------|------------|
| 2 | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி              | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3 | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி             | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை       | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 6 | marketplace | Farmer Market                             | 11.4146849,<br>78.0805463 | 30 km/h     | Medium     |
| 7 | school      | Government Higher Secondary School        | 11.4150045,<br>78.0838172 | 30 km/h     | Medium     |
| 8 | school      | Government school                         | 11.4248027,<br>78.1323902 | 30 km/h     | Medium     |
| 9 | college     | Paavai Arts And Science College For Women | 11.4262443,<br>78.1546987 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38346, 77.89975





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.38302, 77.89995



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38188, 77.90257



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.37841, 77.90140





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.36745, 77.93951



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.41332, 78.08072





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41463, 78.08067



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.42964, 78.11587



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42454, 78.15291







**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44925, 78.22288



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44813, 78.22305



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44793, 78.22320





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49628, 78.32342



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50497, 78.33079





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50769, 78.33291



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50860, 78.33951



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.51180, 78.34422

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